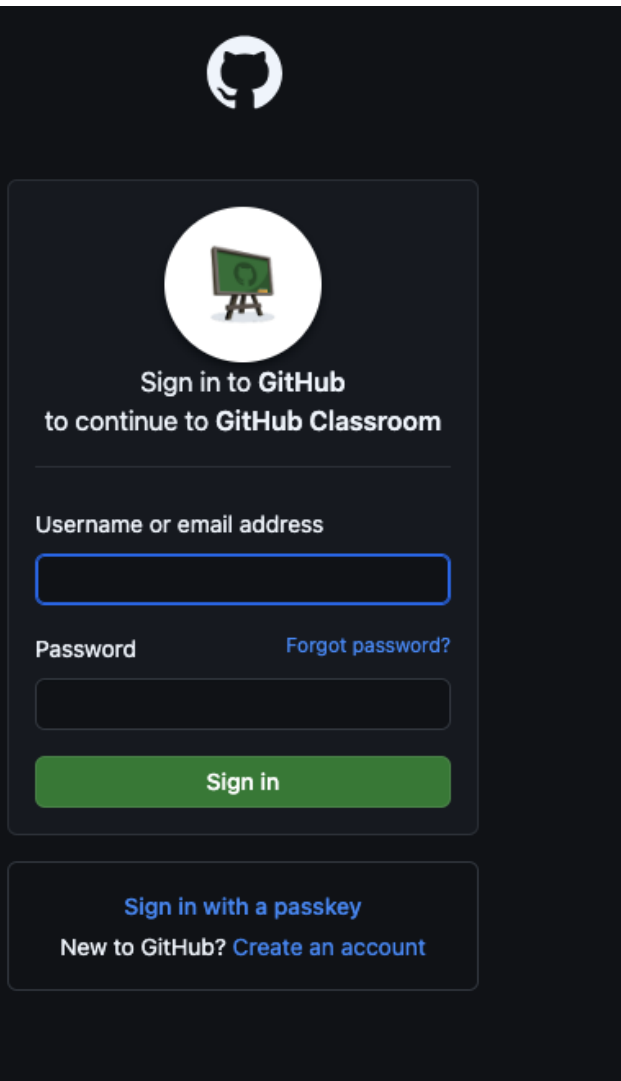


Advanced Programming

Project assignment – 1st stage

Inheritance, Polymorphism, and Abstract Classes

Repository setup



- Students must access the *GitHub Classroom* link provided by the teacher and log in with their *GitHub* accounts
- Initially only one person of each group joins to create the team and then the other students in the group will join

Join the classroom (1 student per group)

- Select your name from the list to join the classroom
- Only students from the corresponding class will show up
- If you can't find your name in the list, report to the teacher

Join the classroom:

Turma P1

To join the GitHub Classroom for this course, please select yourself from the list below to associate your GitHub account with your school's identifier (i.e., your name, ID, or email).

Can't find your name? [Skip to the next step →](#)

Identifiers	
Afonso	>
Ana	>
Carolina	>
Henrique	>
João	>

Creating the team (1 student per group)

Turma P1

Accept the group assignment — Chess Game

Before you can accept this assignment, you must create or join a team. Be sure to select the correct team as you won't be able to change this later.

Join an existing team:

Group 1 0 students	Join	Group2 0 students	Join
--------------------	------	-------------------	------

Or... Create a new team:

+ Create team

- The first student of each group to register will create the team with the provided group name

Group IDs

P1	P2	P3	P4	P5	P6	TE
Group-03	Group-07	Group-15	Group-01	Group-10	Group-02	Group-12
Group-18	Group-11	Group-17	Group-05	Group-14	Group-04	Group-31
Group-21	Group-16	Group-20	Group-06	Group-30	Group-25	Group-39
Group-22	Group-28	Group-23	Group-08	Group-33	Group-32	Group-55
Group-24	Group-29	Group-26	Group-09	Group-46	Group-36	Group-60
Group-27	Group-37	Group-35	Group-13	Group-48	Group-45	Group-65
Group-42	Group-38	Group-40	Group-19	Group-50	Group-49	Group-74
Group-69	Group-41	Group-54	Group-34	Group-56	Group-51	
Group-72	Group-43	Group-58	Group-44	Group-57	Group-79	
Group-73	Group-47	Group-61	Group-52	Group-59	Group-82	
Group-80	Group-62	Group-68	Group-53	Group-63	Group-84	
Group-81	Group-75	Group-70	Group-66	Group-64		
Group-85	Group-76	Group-71	Group-67	Group-78		
	Group-77					
	Group-83					

Accept assignment (1 student per group)

- After accepting the assignment, the success screen will be shown
- Access the repository link and add it to your bookmarks
- Now, the rest of the students in the group can start registering



You're ready to go —

Group 3

You accepted the assignment, **Chess Game**.

Your team's assignment repository has been created:

 <https://github.com/IsecPATestOrg/chess-game-group-3>

We've configured the repository associated with this assignment.

 Your assignment is due by **Jun 1, 2025, 17:26 UTC**

Note: You may receive an email invitation to join [IsecPATestOrg](https://github.com/IsecPATestOrg) on your behalf. No further action is necessary.



Remaining students registration

Turma P1

Accept the group assignment —

Chess Game

Before you can accept this assignment, you must create or join a team. Be sure to select the correct team as you won't be able to change this later.

Join an existing team:

Group 1 0 students	Join	Group2 0 students	Join
--------------------	------	-------------------	------

Or... Create a new team:

+ Create team

- Same process as the first student of the group
- Choose your team from the existing list by clicking **Join**

Advanced Programming – Work assignment

First week

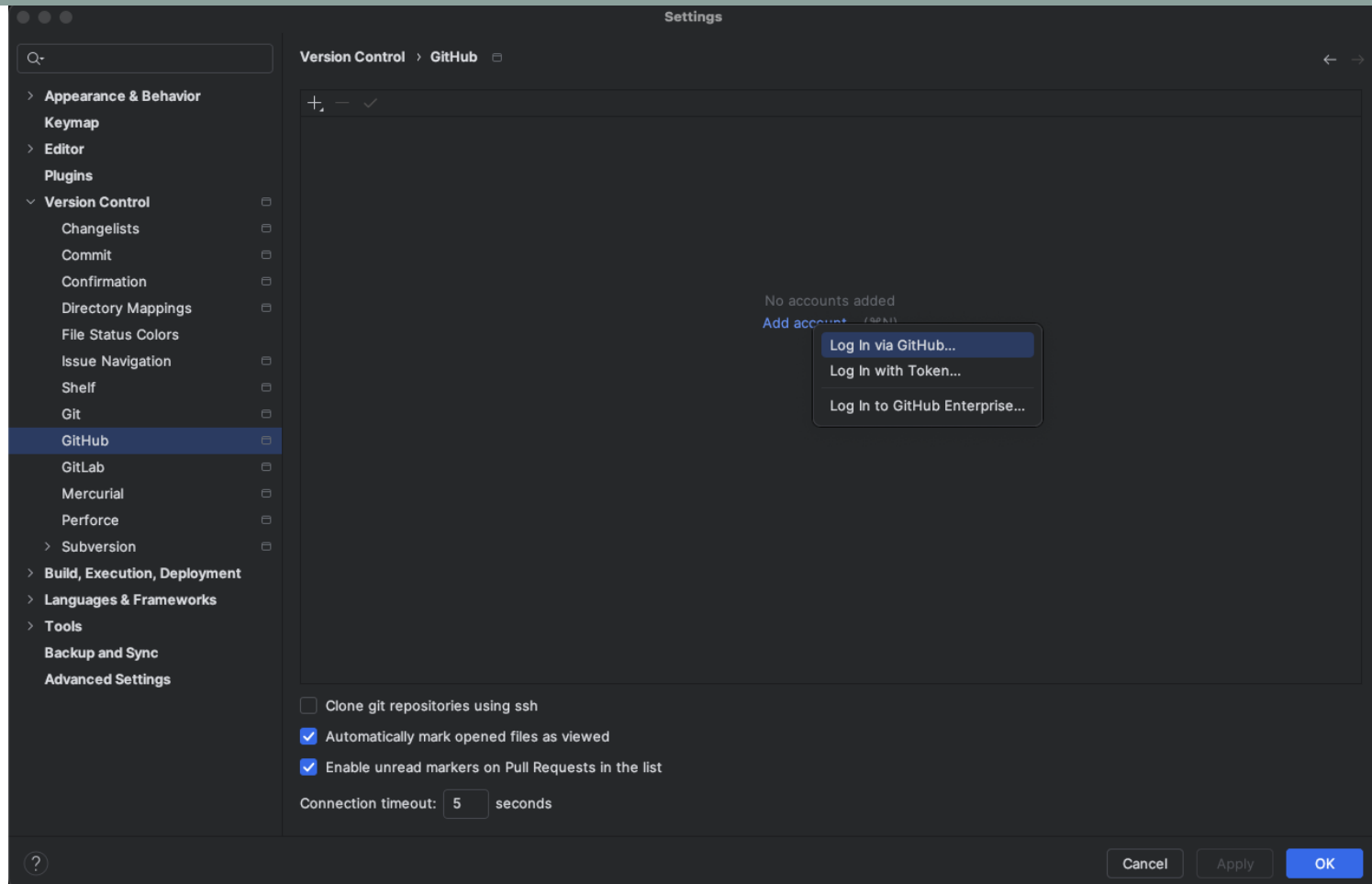
Cloning the repository in IntelliJ



GitHub

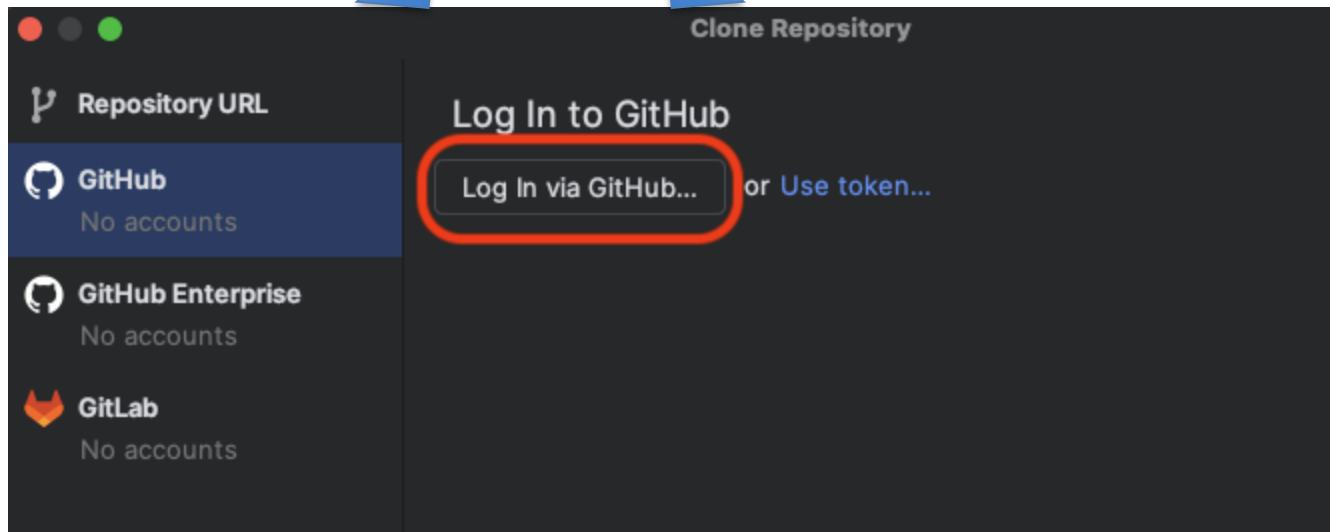
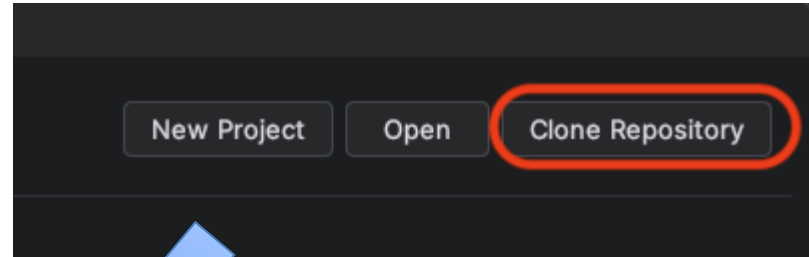
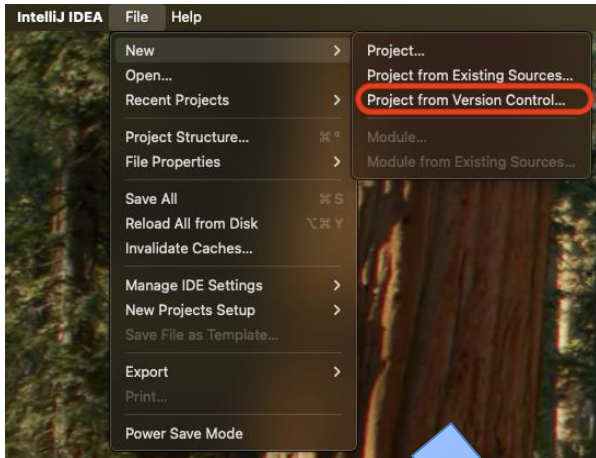


Login into Github

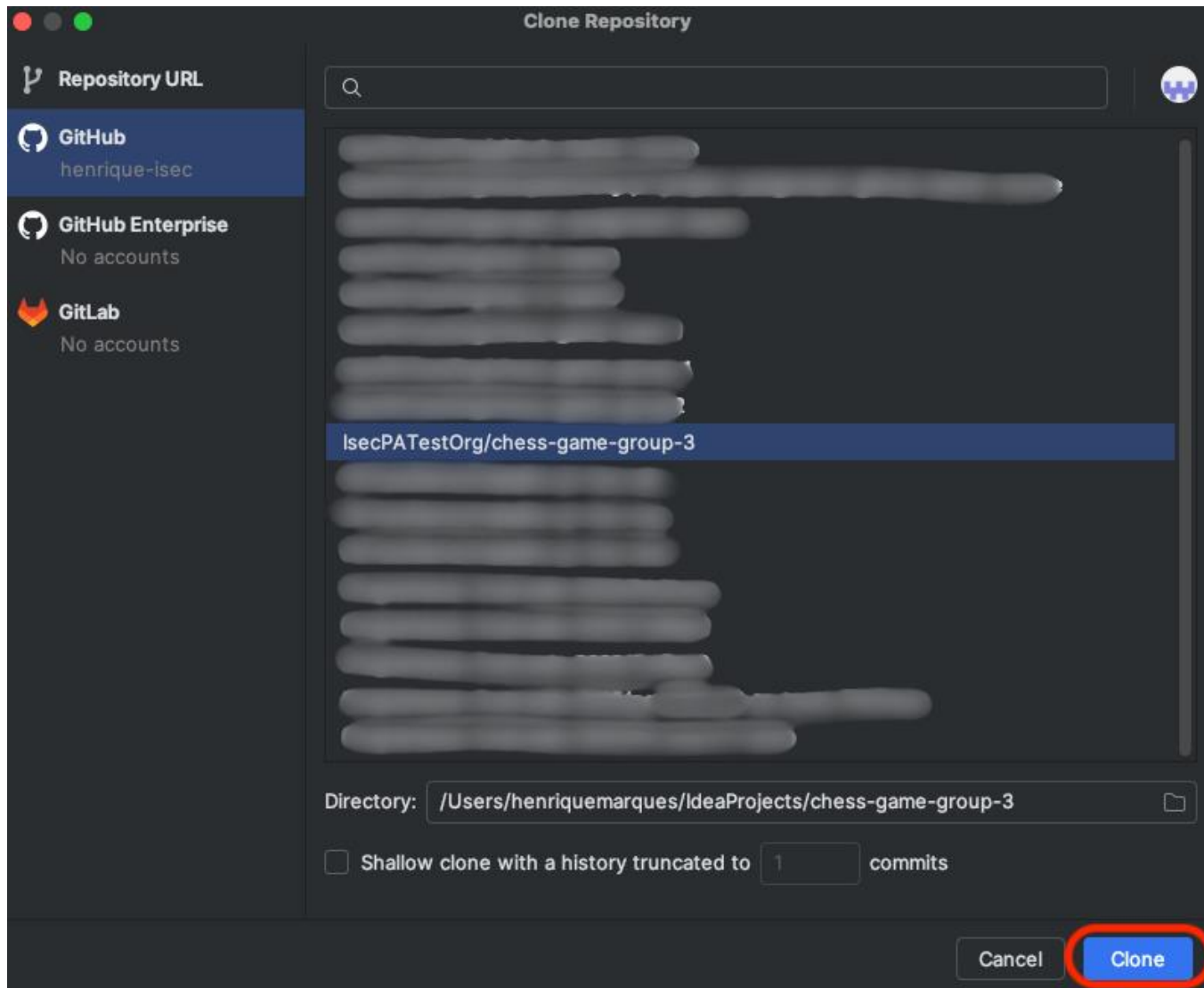


- Go into Settings > Version Control > Github > Add Account > Log In via Github
- Authorize the connection in the browser
- Confirm the account has been added and click Apply

Login into Github (alternative)



Clone the repository



Advanced Programming – Work assignment

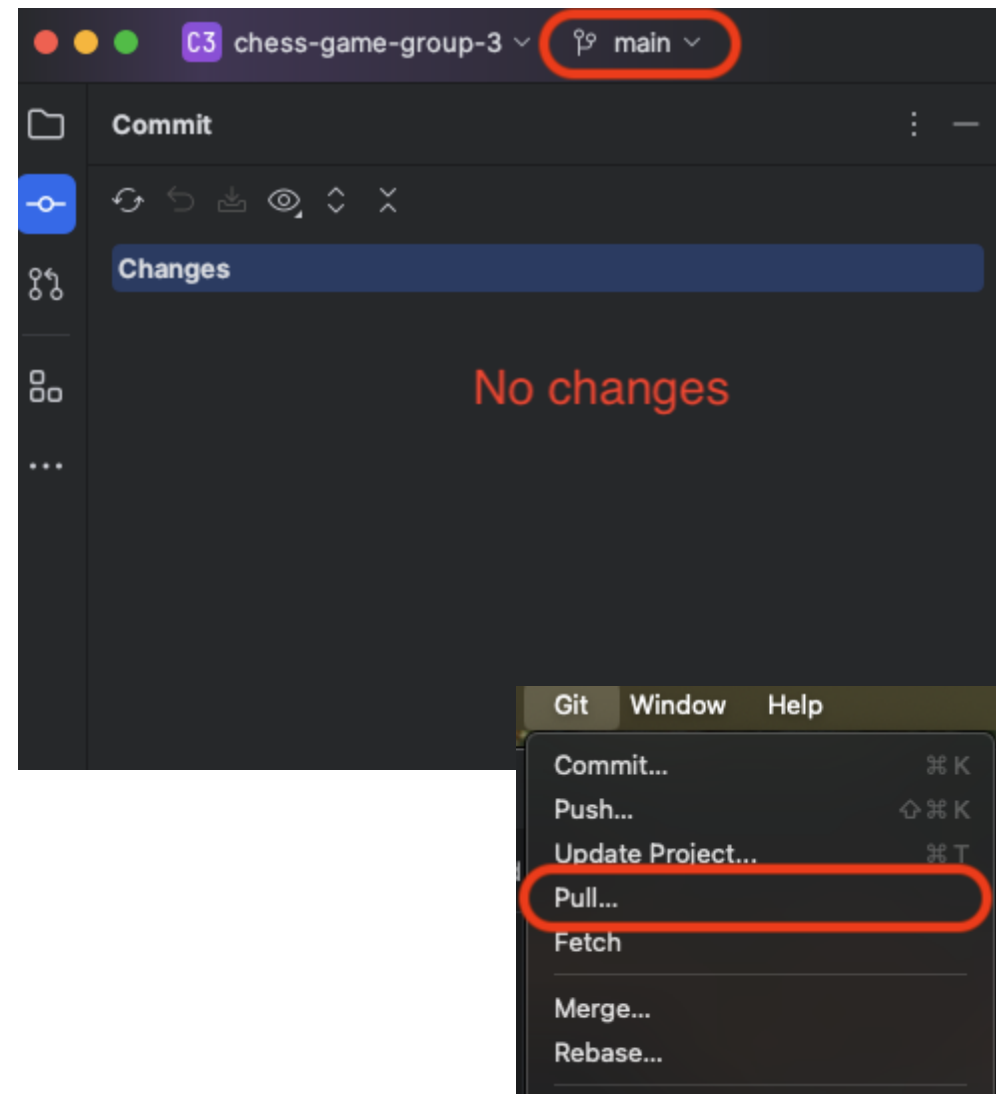
Create a new branch



GitHub

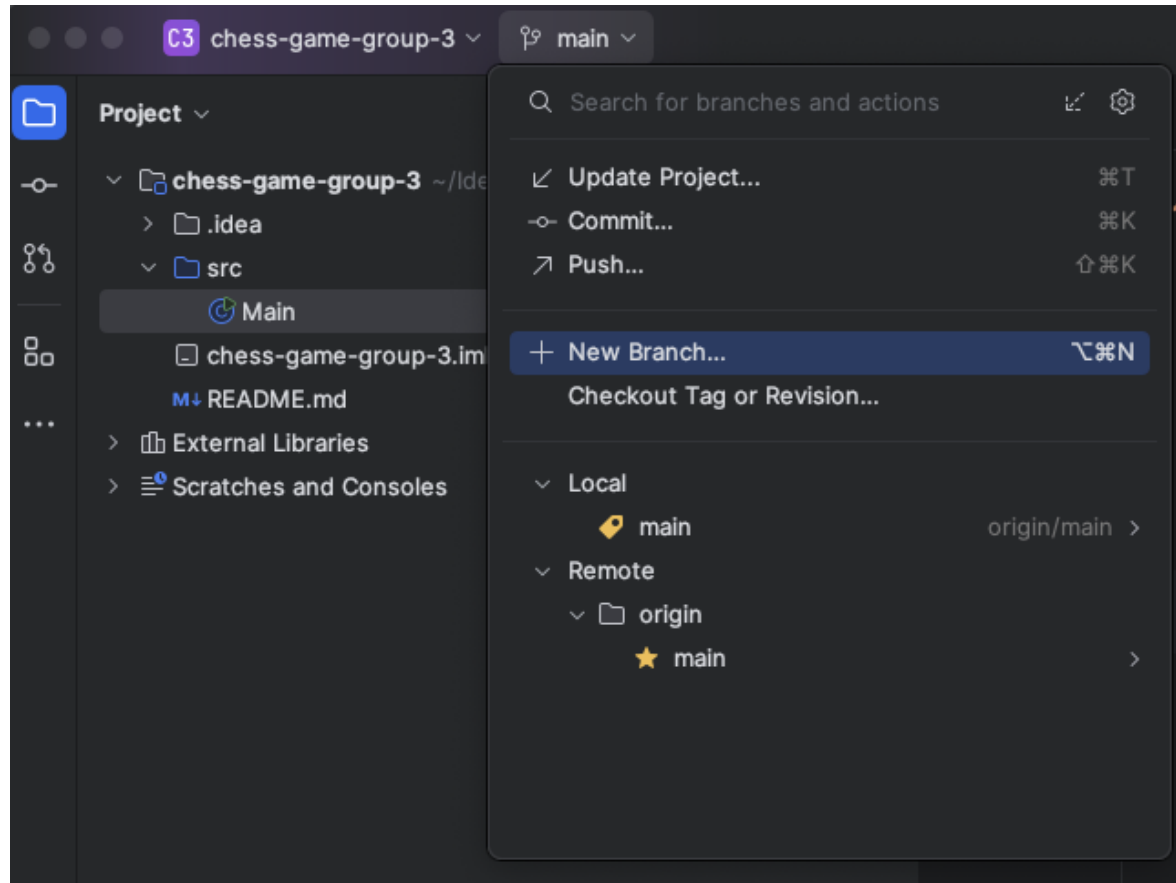


Before create or checkout a new branch



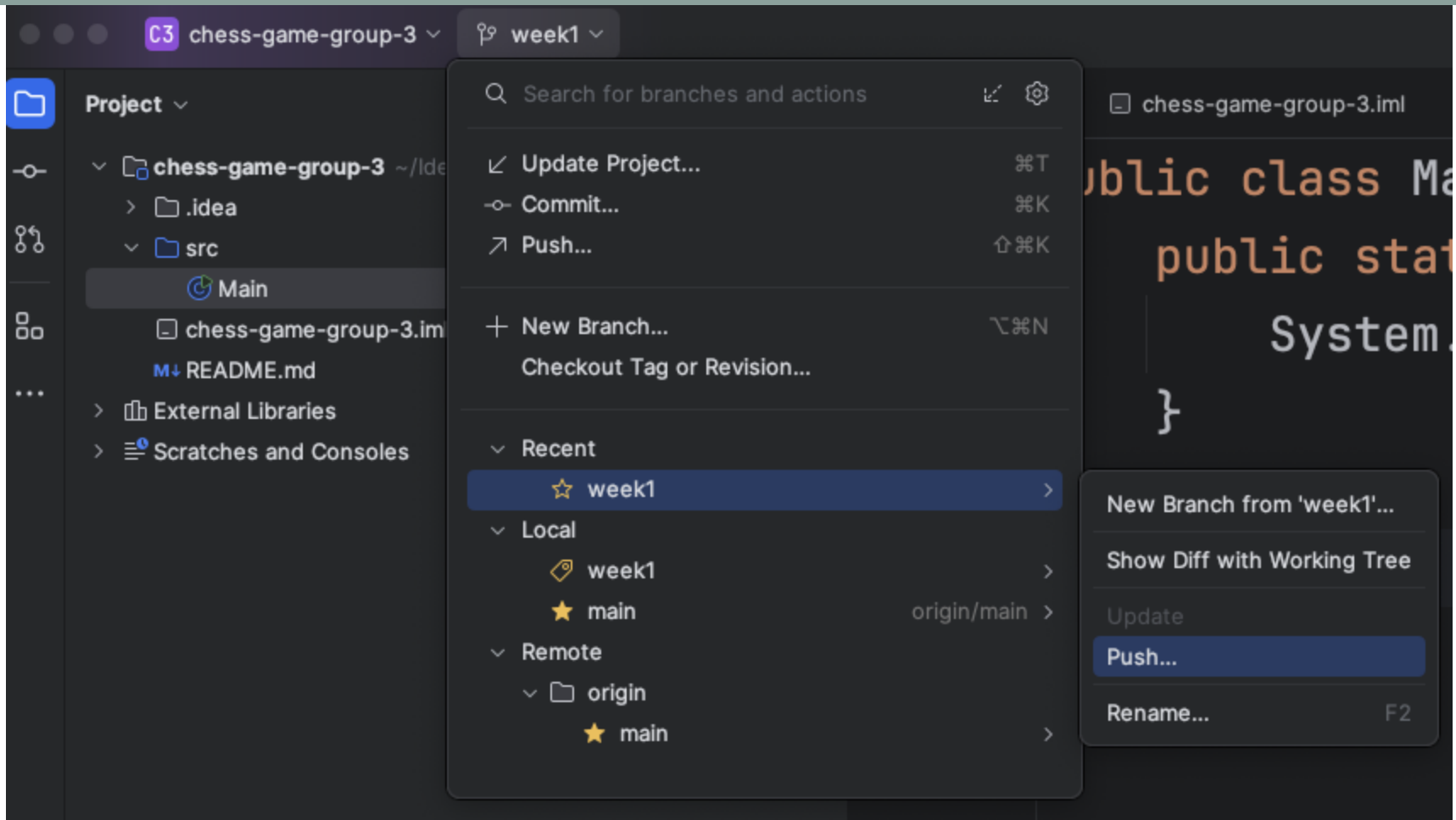
- Make sure that there are no uncommitted changes and that the current branch is main
- Update the main branch by clicking the menu **Git** and then **Pull**

Create the new branch (1 person in the group)



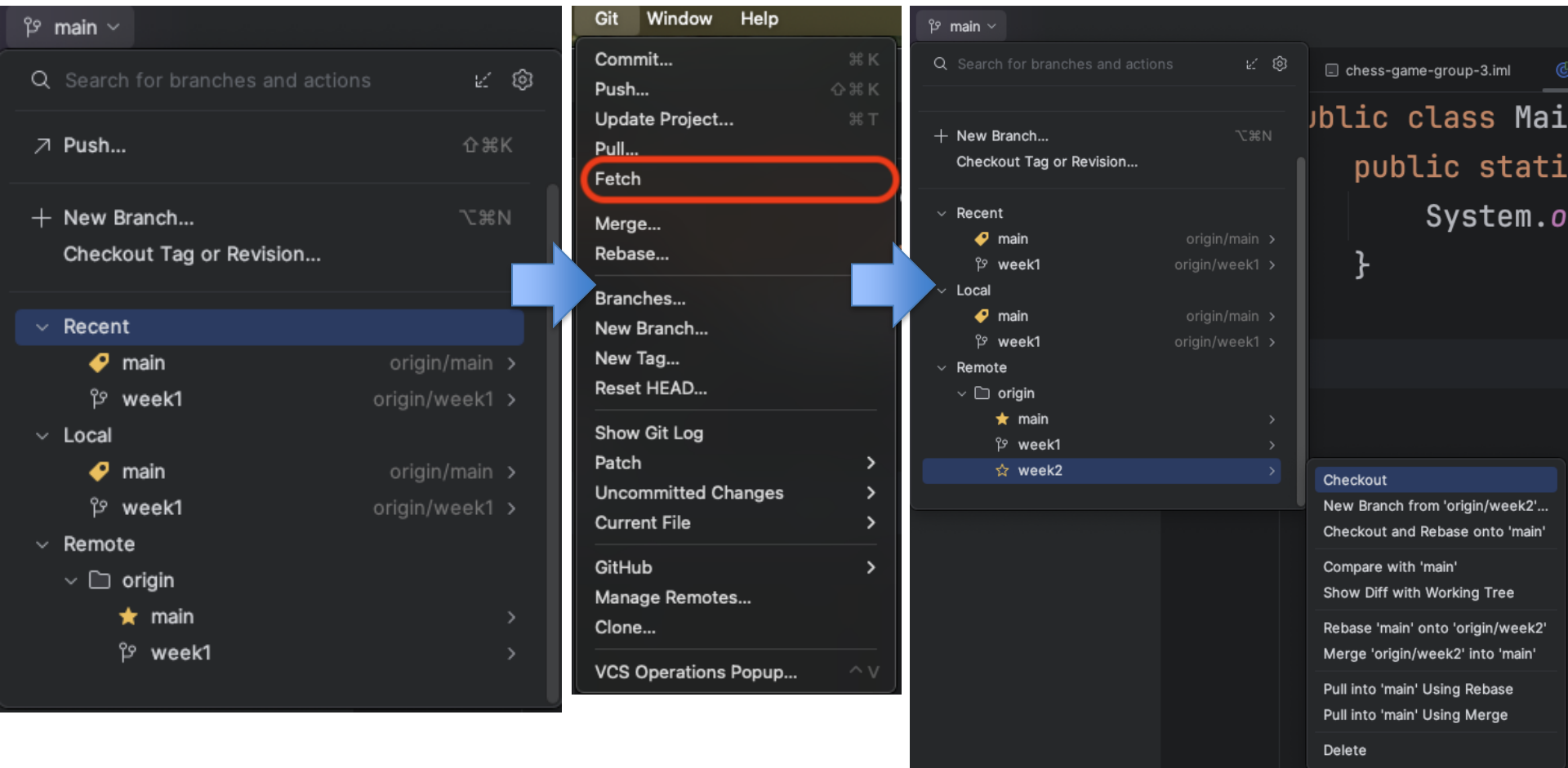
- Branch name: **stage1**

Push the new branch (1 person in the group)



- The new branch must be pushed to be visible to other people in the group

Checkout the new branch (other people in the group)

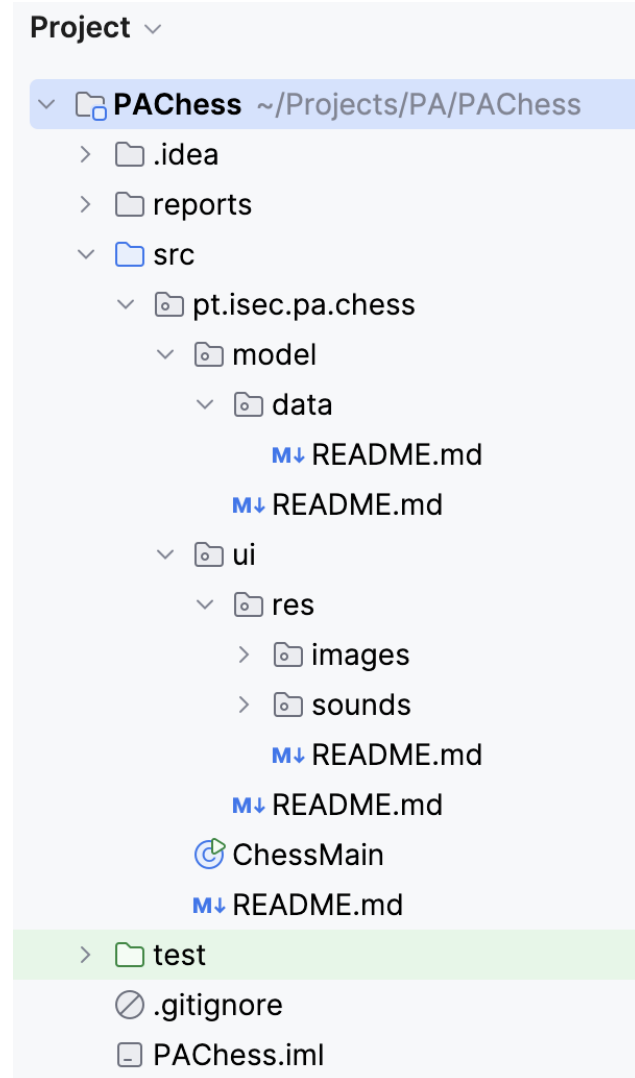


Advanced Programming

Project structure

Project structure

- The provided repository reflects the package structure outlined in the practical work instructions
- Additionally, it includes the following folders:
 - *reports*
 - This folder should contain all reports, diagrams, and supporting files related to the work description
 - Each student should place their step-by-step report in this folder
 - All students will also place their final report here
 - *test*
 - This directory will include the unit tests mentioned in the practical work instructions.



Advanced Programming

PAChess

Stage 1

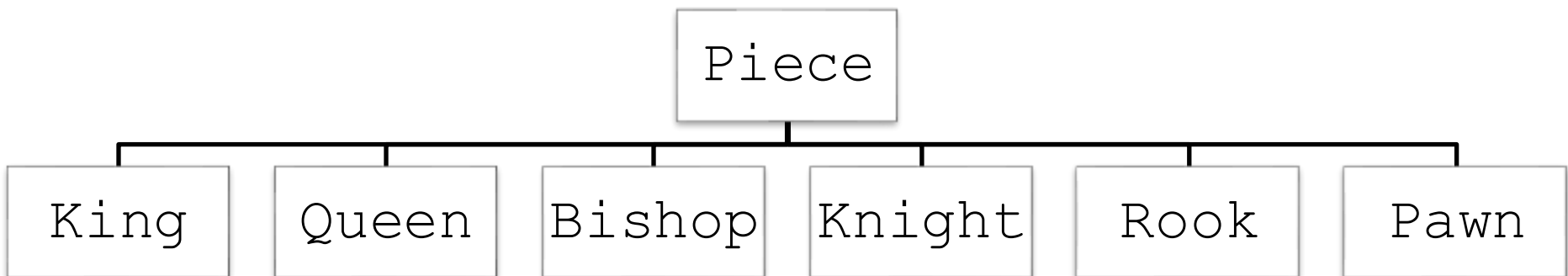
- Define object classes to represent:
 - Game Board
 - Game Pieces
- The class representing the chess game, `ChessGame`, which will manage the progression of the game, will be defined in the next stage

Stage 1

- **Game Board:** `Board`
 - This class should contain a variable to represent the entire game board, ensuring it can store and manage all game elements
 - It should provide methods to add, remove, and manipulate game pieces on the board
 - It should include helper methods to validate game rules

Stage 1

- Game Pieces:
 - Define an abstract class to represent a generic game piece:
`Piece`
 - Define subclasses for each of the chess pieces
 - These classes should include methods to:
 - Return the possible moves for each chess piece
 - Provide normalized text representations for each piece and its position on the board (see next slide)



Stage 1

- Text representations for chess pieces

`<Piece><Column><Row> [ * ]`

- Piece:

- White pieces: K, Q, B, N, R, P
- Black pieces: k, q, b, n, r, p

- Position on the board:

- Column: a to h
- Row: 1 to 8

- An asterisk (*) denotes that the *King* and/or *Rook* have not moved

- Based on these definitions, the initial board can be represented as follows:

```
ra8*,nb8,bc8,qd8,ke8*,bf8,ng8,rh8*,
pa7,pb7,pc7,pd7,pe7,pf7,pg7,ph7,
Pa2,Pb2,Pc2,Pd2,Pe2,Pf2,Pg2,Ph2,
Ra1*,Nb1,Bc1,Qd1,Ke1*,Bf1,Ng1,Rh1*
```